AlpineWatch-1

Phase 0 Mission Proposal

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Executive Summary

The mission begins with launch on a compatible launch vehicle and ascent to the injection conditions specified for the mission orbit. During the launch phase, the spacecraft remains unpowered or in the launch configuration required by the launch service provider. Orbit insertion is performed by the launch vehicle to place the spacecraft into a 550 km circular orbit at 97.6 deg inclination within the allowed injection tolerances; if included in the final design, the spacecraft may perform limited orbit trim or phasing maneuvers after separation. Immediately after separation, the spacecraft transitions autonomously to a post-separation survival mode, stabilizes power, reduces attitude rates, establishes a safe attitude as applicable, and transmits initial health telemetry for ground acquisition. During commissioning, operators verify spacecraft bus functions, communications, power, thermal control, command and data handling, timing, and attitude functions, then activate and check out the payload and obtain at least one first-light image to confirm basic imaging capability. During nominal operations, the ground segment uploads activity plans, the spacecraft performs imaging on commanded daylight passes, stores payload and housekeeping data onboard, and downlinks telemetry and image data during scheduled ground contacts. Routine operations also include orbit determination updates, calibration activities, trend monitoring, and maintenance actions if required by the final design. During contingency operations, predefined faults trigger safe mode, payload operations are suspended, survival power and thermal control are maintained, and the spacecraft awaits ground diagnosis and recovery commands; if available, reduced-function operational modes may be used to recover partial mission capability. At end of life, the spacecraft terminates routine imaging, performs passivation of stored energy sources, executes the approved disposal strategy, and transitions to a non-operational state consistent with applicable debris mitigation requirements.

MVP Scope and Limitations

1. This MVP report is generated for a single-satellite mission concept. It does not analyze constellation deployment, multi-satellite phasing, revisit optimization, inter-satellite links, or multi-plane architectures.
2. This MVP report assumes an Earth Observation payload concept. Payload-specific analysis is limited to high-level swath, resolution, mass, and power assumptions where provided.

Mission Overview

- Acquire multispectral Earth observation imagery from a 550 km circular, 97.6 deg inclination orbit with 5 m ground sample distance at nadir during the mission lifetime.
- Provide a minimum imaged swath width of 40 km at nadir for each payload acquisition.
- Operate the spacecraft and payload for at least 3.0 years after completion of on-orbit commissioning.
- Generate, store, and downlink payload data and housekeeping telemetry required for routine Earth observation operations and mission performance assessment.
- Commission the spacecraft and payload after launch and demonstrate on-orbit capability for nominal imaging operations before routine operations begin.

Mission Requirements

ID	Requirement	Rationale
MR-01	The mission orbit shall be 550 km circular altitude ± 25 km with an inclination of 97.6 deg ± 0.2 deg during nominal [illegible]	The operational orbit defines observing geometry, [illegible]

ID	Requirement	Rationale
MR-02	The mission shall provide an operational lifetime of at least 3.0 years from completion of on-orbit commissioning.	The mission duration is a primary program requirement
MR-03	The payload shall acquire multispectral imagery with a ground sample distance of 5 m or finer at nadir from the nomina	Ground sample distance is a primary payload performa
MR-04	The payload shall provide an imaged swath width of at least 40 km at nadir from the nominal operational orbit.	Swath width defines the area coverage per acquisition.
MR-05	The spacecraft shall provide at least 35 W of regulated electrical power to the payload during imaging operatio	The payload power demand must be supported durin
MR-06	The electrical power subsystem shall provide electrical power for spacecraft housekeeping loads and payload imaging	Mission operations require support of platform and payl
MR-07	The spacecraft shall support commanded payload imaging during sunlit orbital passes.	Multispectral Earth observation imaging requires solar i
MR-08	The spacecraft shall time-tag all payload image data and housekeeping telemetry using a single onboard time reference	A common time reference is required for data correlatio
MR-09	The command and data handling subsystem shall store payload data and housekeeping telemetry	Onboard storage is required to bridge periods without g
MR-10	The communications subsystem shall provide telemetry, tracking, and command capability for spacecraft health monito	Ground interaction is required to operate the spacecraf
MR-11	The communications subsystem shall provide a payload data downlink capability throughout the mission lifetime.	Payload data return is required to achieve the mission o
MR-12	The spacecraft shall autonomously enter a post-separation mode that performs initial power stabilization, attitude rate	These functions are required to establish a post-launch s
MR-13	The spacecraft shall complete on-orbit commissioning before entry into routine payload operations.	Commissioning is required to verify readiness for susta
MR-14	The spacecraft shall provide a safe mode that inhibits payload operations and maintains spacecraft power and	Autonomous safing preserves spacecraft survival when
MR-15	The mission shall execute an end-of-life disposal sequence that passivates stored energy sources and terminates	Passivation and operational termination are required
MR-16	The spacecraft shall provide orbit state data for each payload acquisition.	Orbit knowledge is required for image geolocation proc
MR-17	The thermal control subsystem shall maintain the payload within its qualified operating temperature limits during	Payload operation depends on maintaining temperatu
MR-18	The structures subsystem shall withstand the quasi-static, dynamic, and acoustic launch environments specified by the	The spacecraft must survive the launch environment
MR-19	The spacecraft shall transmit health telemetry within 30 minutes after separation from the launch vehicle.	A bounded initial contact interval is required for early o
MR-20	The spacecraft shall demonstrate at least one payload image acquisition during on-orbit commissioning.	A first-light acquisition is a verifiable indicator of payloa

Mass Budget

Status: YELLOW

Subsystem	Mass (kg)	Margin %	Notes
Structure	10.8	20.0	Estimated primary and secondary structure for a small EO spac
ADCS	8.4	20.0	Includes sensors, wheels, magnetorquers, and GN&C hardware
Propulsion	2.4	20.0	Minimal allowance for optional orbit trim, momentum manageme
Communications	4.8	20.0	Includes TT&C plus payload data downlink hardware sized at a
OBC/Data Handling	4.2	20.0	Includes flight computer, mass memory, timing, interfaces, and
EPS/Power	10.2	20.0	Includes solar arrays, batteries, power regulation/distribution, ar
Thermal	3.6	20.0	Passive thermal control with heaters, MLI, coatings, and therma
Payload	9.6	20.0	Based on stated payload mass of 8 kg with 20% subsystem ma
Harness	2.4	20.0	Preliminary wiring and connector allowance consistent with a sn

Power Budget

Status: GREEN

Mode	Total W

Cost Estimate

Category	Cost (kEUR)	Basis
Spacecraft Hardware	5200.0	Engineering judgment
Integration & Test	1700.0	Engineering judgment
Launch	2800.0	Engineering judgment
Ground Segment	900.0	Engineering judgment
Operations	1500.0	Engineering judgment
Program Management	1500.0	Engineering judgment
Contingency	2720.0	Engineering judgment

Component Alternatives

Assumptions and Open Items

Assumption: The launch service provider will inject the spacecraft into an orbit within ±25 km altitude and ±0.2 deg inclination of the nominal mission orbit; if this assumption is wrong, onboard propulsion delta-v requirements may change materially.
Assumption: The 5 m ground sample distance and 40 km swath requirements apply at nadir from the nominal operational orbit; if off-nadir performance is required, payload aperture, focal plane, and attitude control sizing may change materially.
Assumption: The payload requires at least 35 W regulated power during nominal imaging operations; if peak or average power is higher, solar array and battery sizing may change materially.
Assumption: Routine imaging is limited to sunlit orbital passes; if twilight or night imaging is required, payload design and operations concept may change materially.
Assumption: A ground segment with sufficient contact opportunities and payload data throughput will be available; if not, onboard storage capacity and downlink rate requirements may increase materially.
Assumption: The mission does not require continuous propulsion for nominal operations; if propulsion is required for orbit maintenance, collision avoidance, or disposal, spacecraft mass, volume, power, and safety constraints may change materially.
Assumption: Passive thermal control plus operational heaters can maintain all units within qualified temperature limits in the 550 km orbit; if not, radiator area, heater power, and thermal hardware mass may change materially.
Assumption: The selected orbit provides illumination conditions compatible with routine multispectral imaging; if local time of descending node or seasonal lighting differs from this assumption, mission utility and imaging opportunities may change materially.
Assumption: The spacecraft can meet debris mitigation obligations either through natural decay from 550 km or through a dedicated disposal function; if this assumption is wrong, additional propulsion or drag augmentation hardware may be required.
Assumption: Payload data generation rate, onboard storage requirement, and downlink volume have not yet been fixed; if final values are larger than currently assumed, avionics, communications, and power sizing may change materially.
Open question: What spectral bands, band center wavelengths, and minimum signal-to-noise ratio values are required for the payload, expressed per band at a defined reference scene and illumination condition?
Open question: What maximum geolocation error is required for delivered imagery, stated separately for performance with and without ground control points?
Open question: What minimum revisit interval or area coverage rate is required for the target user set, stated for a defined latitude band and cloud-free condition?
Open question: What minimum daily payload data volume shall the mission downlink, in Gbit/day, and what maximum allowable latency shall apply from acquisition to data delivery to users?
Open question: What minimum onboard payload data storage capacity is required, in Gbit, based on the expected imaging plan and worst-case ground contact outage duration?

Open question: What pointing knowledge, pointing control error, and line-of-sight jitter limits are required during imaging to support the 5 m ground sample distance objective?
Open question: What launch vehicle accommodation constraints will apply, including maximum wet mass, stowed envelope, separation interface standard, and injection accuracy bounds?
Open question: Is collision avoidance capability required, and if so, what response timeline, minimum probability-of-collision threshold, and delta-v capability shall be allocated?
Open question: What end-of-life disposal requirement applies, including maximum allowed post-mission orbital lifetime and any mandated passivation actions under the governing authority?
Open question: How many ground stations or relay services will be available, and what minimum contact time per day and minimum downlink data rate will each provide?
Open question: What autonomy level is required for fault detection, isolation, and recovery, including the list of faults that must trigger autonomous safe mode and the maximum allowable time to safe the spacecraft?
Open question: What calibration concept is required to maintain radiometric and geometric performance over 3.0 years, including onboard calibration hardware, lunar calibration, vicarious calibration, or ground-reference campaigns?

Disclaimer

This document is an AI-assisted Phase 0 concept draft. All technical values, cost estimates, procurement options, and requirements must be reviewed and validated by a qualified spacecraft systems engineer before use in design, procurement, proposal submission, or mission decision-making.